

Exam Scheduling System — Validation Checklist

System Overview

The system allows departments and course coordinators to allocate hall locations and time slots for their exams. Its purpose is to minimize scheduling conflicts and keep students updated on available slots, streamlining the exam-setup process.

Coordinator: allocates a location and time, and informs students.

Location: a hall reserved by a coordinator for a specific time slot.

Student: receives exam information from coordinators for enrolled courses.

Components

Three components were selected for validation: Coordinator, Location, and Student. Each entity's fields, validation rules, and corresponding Spring Boot (Bean Validation / Hibernate Validator) annotations are listed below.

1. Coordinator

Variable	Validation Rule
name	Must not be empty; maximum length 30 characters
id	Must not be empty; exactly 11 characters; starts with 'f' followed by 10 digits
course	Must not be empty; letters, numbers, and spaces only

Code nottations:

```
@NotEmpty(message = "Coordinator name is required")
@Size(max = 30, message = "Coordinator name must not exceed 30 characters")
private String name;

@NotEmpty(message = "Coordinator ID is required")
@Pattern(regexp = "^f\\d{10}$", message = "ID must start with 'f' followed by 10 digits")
private String id;

@NotEmpty(message = "Course is required")
@Pattern(regexp = "^[a-zA-Z0-9 ]+$", message = "Course must contain only letters, numbers, and spaces")
private String course;
```

2. Place

Variable	Validation Rule
address	Must not be empty; letters, numbers, and hyphens only (e.g. Building A-101)
timeSlot	Must not be empty; must follow the pattern hh:mm AM/PM
reserved	Must not be null (boolean flag)

Code notations:

```
@NotEmpty(message = "Address is required")
@Pattern(regexp = "[a-zA-Z0-9-]+$", message = "Address must contain only letters, numbers, and hyphens")
private String address;

@NotEmpty(message = "Time slot is required")
@Pattern(regexp = "(0?[1-9]|1[0-2]):[0-5][0-9] (AM|PM)$", message = "Time slot must be in the format hh:mm AM/PM")
private String timeSlot;

@NotNull(message = "Reservation status must be specified")
private Boolean reserved;
```

3. Student

Variable	Validation Rule
id	Must not be empty; exactly 11 characters; starts with 's' followed by 10 digits
courses	Must not be empty (list); each element contains letters, numbers, and spaces only
messages	Must not be null (list of strings)

Code notations:

```
@NotEmpty(message = "Student ID is required")
@Pattern(regexp = "^s\\d{10}$", message = "ID must start with 's' followed by 10 digits")
private String id;

@NotEmpty(message = "At least one course is required")
private ArrayList<@Pattern(regexp = "[a-zA-Z0-9 ]+$",
    message = "Each course must contain only letters, numbers, and spaces") String> courses;

@NotNull(message = "Messages list must not be null")
private ArrayList<String> messages;
```

Note

Applying a constraint directly to a List's element type (e.g. `List<@Pattern(...) String>`) requires Bean Validation 2.0+ container element constraints, supported by Hibernate Validator in Spring Boot 2.x and later.

String fields use `@NotEmpty` (rejects null and empty string) rather than `@NotBlank` (which also rejects whitespace-only values), per project convention. Non-String fields use `@NotNull`.